

Introduction to E-values

(Ramdas & Weng 2025)

Motivation:

$$\boxed{\text{Game}} \xrightarrow{\text{iid}} X_1, X_2, \dots \in [0, 1] \text{ a.s.}$$

Goal - Estimate & do inference on $\mathbb{E}_P(X_i)$.

$$\mathcal{P}_m = \{P: \mathbb{E}_P(X_i) = m\}$$

$$\mathcal{Q} = \{Q: \mathbb{E}_Q(X_i) > m\}$$

Define $W_n = \prod_{i=1}^n (1 + \lambda_i (X_i - m))$

Example of an e-value.

An E-process

Questions

- ① What do they do? Applications.
- ② Properties of E-values (Think $P \stackrel{P}{\leq} U(0,1)$)

This talk does 3 things

- ① Terminology & Definitions
- ② KL Duality & Sequential Testing
- ③ Optimal E-process

- FDR
- E-closure
Not covered.

Definitions A non-negative test statistic whose expected value under the null is at most 1.
 $T \geq 0$ such that $E_P(T) \leq 1$.

What does "e" mean? Evidence, expectation...

Example : Likelihood Ratio : $\frac{dQ}{dP}(x)$

Averages + Products ⁽¹¹⁾ are E-values.

E-values generalize LR to Q, P (composite, non-pure)

Log Optimal $E_Q(\log(\frac{E'}{E})) \leq 0 \quad \forall E'$,

$\Leftrightarrow E$ is log optimal

Use Case : Optional Continuation

Recall the game. We collect 100 data points

$W_{100} = 12$. At 0.05 we can't reject

$P(W_{100} \geq 12) \leq \frac{1}{12}$. Collect 100 more

$W_{100,200} = 4$. $W_{200} = 48$

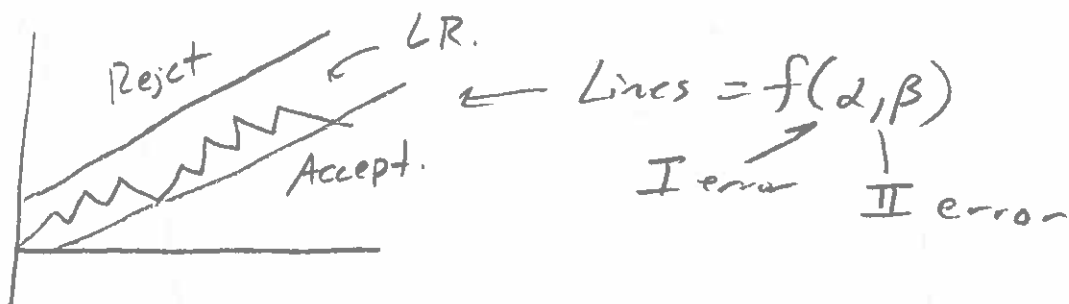
$P(W_{200} \geq 48) \leq \frac{1}{48}$: Martingale.

Sequential Testing

Wald 1945: SPRT (Sequential Probability Ratio Test).

Uses the fact that LR is an e-process.

Picture



Theorem. τ stopping time.

We should always do sequential inference.

$$\max \{ \mathbb{E}_p(\tau), \mathbb{E}_q(\tau) \} \leq n(\alpha, \beta) \leftarrow \text{Stat 101 Power Calc.}$$

May get unlucky



More definitions

Test super M_b , a super M_b (M_t) with the property $\mathbb{E}_p(M_t) \leq 1$.

E-process (E_t) sequence that either

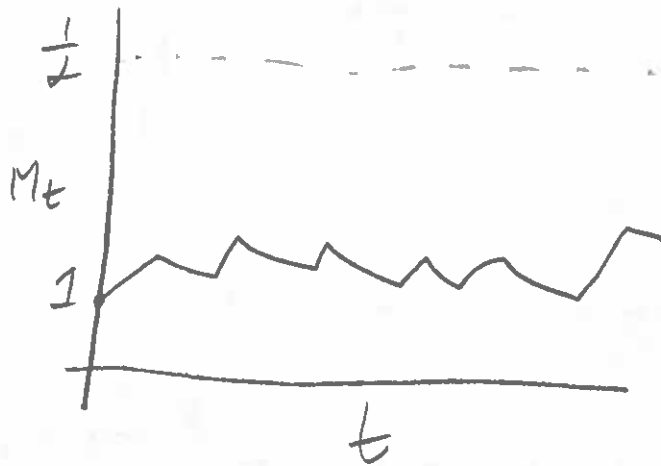
Next Page $\mathbb{E}_p(E_\tau) \leq 1 \forall \tau, - \exists M_t, E_t \leq M_t$

Ville's Inequality. Let (M_t) be a non-negative super-MG

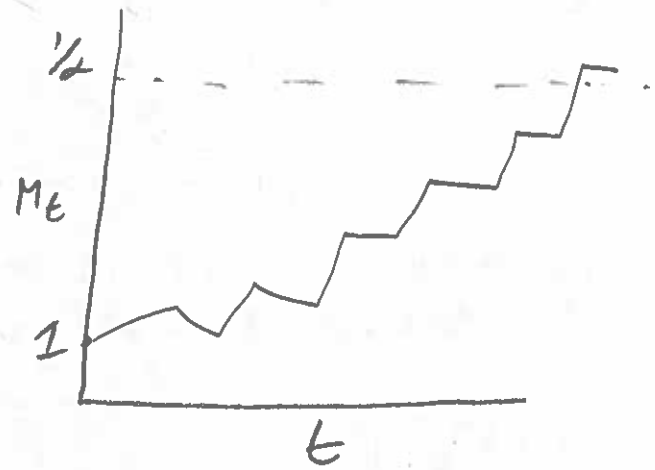
$$P(\exists t: M_t \geq \frac{1}{\alpha}) \leq \frac{E M_0}{\frac{1}{\alpha}} \cdot \alpha \leq \alpha$$

Picture

Fail to Reject

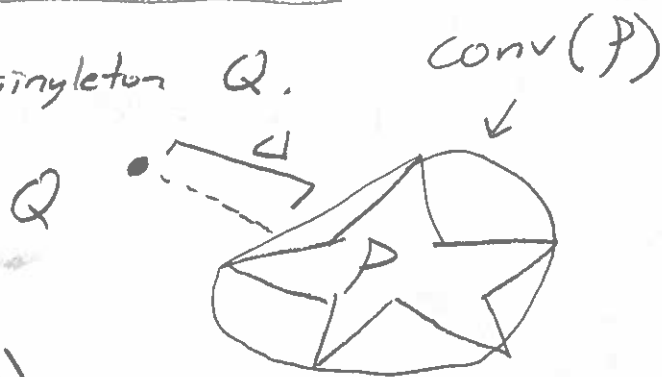


Reject



Duality with KL Divergence

Think composite $\mathcal{P}$, singleton Q .



$$\Delta = KL(Q, \text{conv}(\mathcal{P}))$$

$$= \mathbb{E}_Q(\log E^*), \text{ where } E^* \text{ is the "numeraire".}$$

Formally, $\mathbb{E}_Q(\log E^*) = \max_E \mathbb{E}_Q(\log E) = \min_{P \in \text{conv}(\mathcal{P})} KL(Q, P)$

Example LR is a numeraire.

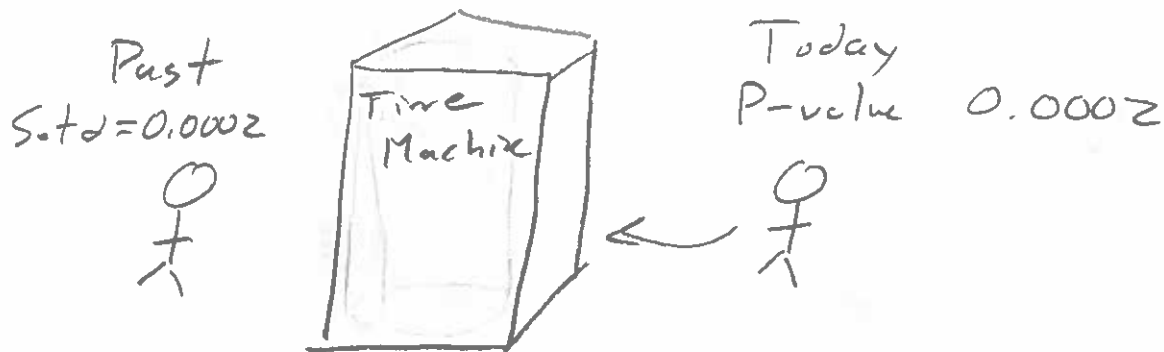
$$E = \frac{dQ}{dP}, \quad \mathbb{E}_Q(\log \frac{dQ}{dP}) \triangleq KL(Q, P).$$

Open question: does this duality extend to
 Q, P . Can we interchange?

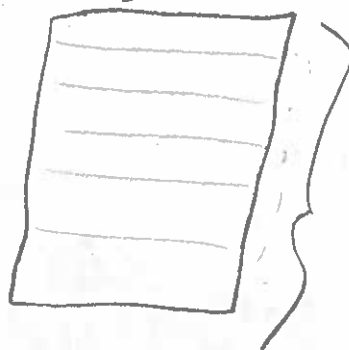
Optimality Think (P, Q)

Neyman-Pearson Optimality

- Maximize Power



- Strange desiderata. Instead think dataset



N , instead say use $0.6 \cdot N$

Note equivalent in general. Different.

Universal Optimality: Asymptotic so Distributions "Flatten"

Easy for P, Q, hard for $\mathcal{P}, \mathcal{Q}$

Recent Papers

(Woodby-Smith, Sundaram, Jordan) 2025

(Agrawal, Ramdas) 2025

Deals with questions of $KL \cong \mathbb{E}_Q(\log E^*)$.

A Axiom, all problems are "betting" problems

$$W_n = \prod_{i=1}^n (1 + \lambda_i (E_i - 1))$$

Optimall $\Leftrightarrow$ Optimall (λ_n) .

Def $\tau_\alpha = \inf \{n: W_n \geq 1/\alpha\}$. want to

$$E_Q(\tau_\alpha).$$

Result: Choose (λ_n) to maximize compound growth.
 $\alpha \rightarrow 0$.

$$\frac{\log(1/\alpha)}{l_Q(\lambda)} \leq E_Q(\tau_\alpha) \leq \frac{\log(1/\alpha)}{l_Q(\lambda)}.$$

$$l_Q(\lambda) = E_Q(\log(1 + \lambda(E - 1)))$$

Popular algorithms like Online Newton Step
don't guarantee this.

Agarwal Approach

① $\alpha \rightarrow 0$ some except define $KL(Q, P)$

as $\inf \{ Q \in \mathcal{Q} \mid KL(Q, P) \} \cong l_{\mathcal{Q}}^{\star}(1)$

(subset of problems)

② Define $\Delta_n = KL(Q_n, P)$. Take $\Delta \rightarrow 0$.

$$\text{get } \lim_{n \rightarrow \infty} \frac{\mathbb{E} Q_n(\tau_{\Delta})}{\Delta n^{-1} \log \log \Delta n^{-1}} = O(1).$$

Strange Structure. LIL has $\sqrt{n \log \log n}$ and is in Probability rather than expectation.

"Modern Result"

Farrell (64)

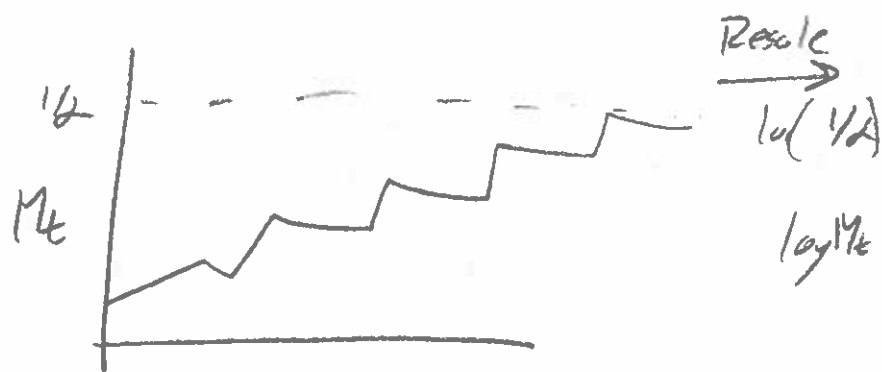
$$\lim_{\theta \rightarrow 0} \frac{\mathbb{E} \theta(\tau_{\Delta})}{\theta^{-2} \log \log \theta^{-1}} = O(1).$$

for exponential families.

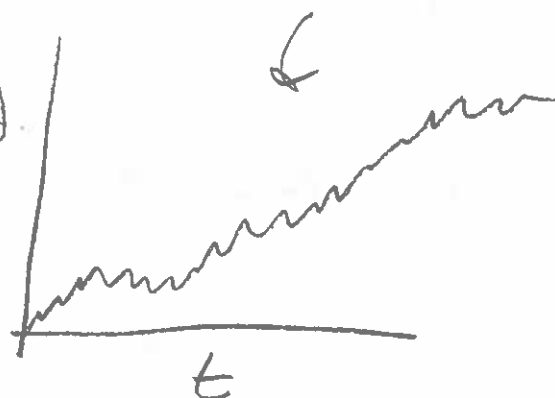
Their method of showing this

$$\lim_{n \rightarrow \infty} \frac{KL_{\inf}(Q_n, P)}{l_{\mathcal{Q}_n}^{\star}} = 1$$

I'm thinking about



Brownian Motion



3rd Regime $n \rightarrow \infty$ $\alpha = \alpha_0$

$T_2 \xrightarrow{n \rightarrow \infty} \text{Inv-Gaussian} \left(\frac{\log(1/\alpha)}{\log(\lambda)}, \frac{\log(1/\alpha)^2}{\sqrt{\text{Variance}}} \right)$

Recover Previous Results.

Can you prove the general?

$$\lim_{n \rightarrow \infty} \frac{KLinf(Q_n, P)}{\log(\dots)} = 1.$$